

CARDIAC CYCLE

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- **Definition** : Succession of (sequence of) coordinated mechanical and electrical events taking place in the heart during each beat.
- **Comprises 2 Major Periods** :—
 - **Systole** : Period during which heart muscle contracts.
 - **Diastole** : Period during which heart muscle relaxes.
- These changes repeat cyclically with every heartbeat.

DIVISIONS AND DURATION OF CARDIAC CYCLE

Normal Heart Rate = 72 beats/min $\implies$ Duration of each Cardiac Cycle = $\frac{60}{72} = 0.8$ seconds

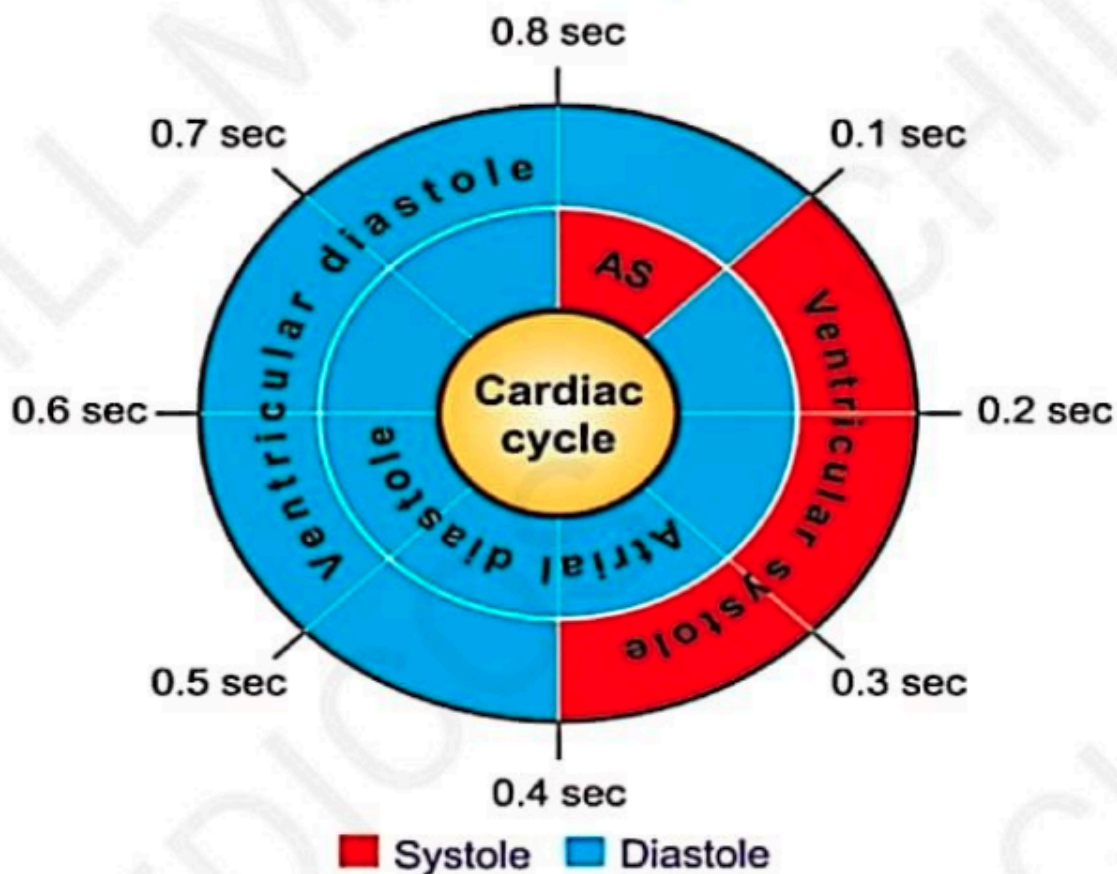


FIGURE 91.1: Atrial and ventricular events of cardiac cycle

Division	Systole Duration	Diastole Duration	Total Duration
Atrial Events	0.1 second	0.7 second	0.8 second
Ventricular Events	0.3 second (0.27 s)	0.5 second (0.53 s)	0.8 second

Complete Cardiac Diastole : The heart relaxes as a whole (both atria and ventricles in diastole simultaneously) for **0.4 seconds**.

1. ATRIAL EVENTS

• A. Atrial Systole (0.1 sec) :—

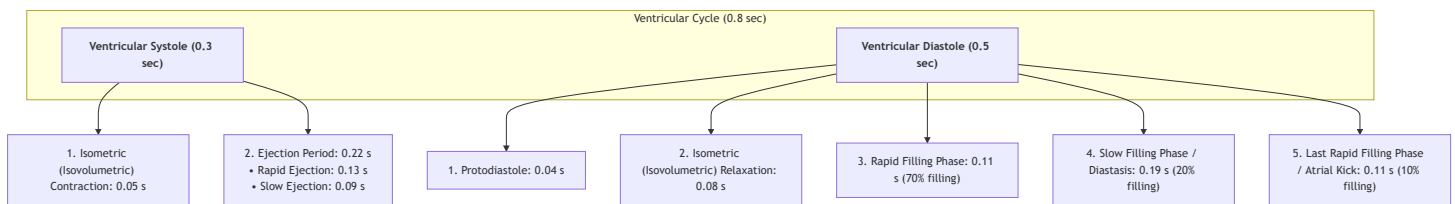
- Also known as the **last rapid filling phase** or **presystole** of ventricular diastole.
- *Functional Significance:* Not strictly essential for maintenance of basic circulation (a person with atrial fibrillation can survive for years without developing severe circulatory insufficiency).
- **Pressure & Volume Changes:** Intra-atrial pressure ↑ increases; intraventricular pressure and volume also ↑ increase slightly.

- **Fourth Heart Sound (S_4):** Caused by the contraction of atrial musculature and rapid ventricular inflow.
- **B. Atrial Diastole (0.7 sec) :—**
 - Starts simultaneously as ventricular systole begins.
 - **Necessity of Long Atrial Diastole:** Provides adequate filling time:

Right Atrium receives deoxygenated blood via Superior & Inferior Venae Cavae

Left Atrium receives oxygenated blood from lungs via Pulmonary Veins

2. VENTRICULAR EVENTS



A. PHASES OF VENTRICULAR SYSTOLE (0.3 sec)

- **1. Isometric / Isovolumetric Contraction Period (0.05 sec) :—**
 - First phase of ventricular systole.
 - Type of muscular contraction characterized by $\uparrow$ **increase in tension without change in length of muscle fibers**.
 - **Valvular Status:**
 - After atrial systole, **AV valves close** due to rising intraventricular pressure.
 - Semilunar valves are **already closed**.
 - Ventricles contract as **closed cavities** $\longrightarrow$ intraventricular volume remains constant while intraventricular pressure spikes steeply.
 - **First Heart Sound (S_1):** Produced by the closure of AV valves (mitral and tricuspid) at the beginning of this phase.
 - **Significance:** Once intraventricular pressure exceeds aortic/pulmonary pressure, **semilunar valves open**.
- **2. Ejection Period (0.22 sec) :—**
 - **Rapid Ejection Period (1st stage, 0.13 sec):** Large amount of blood is rapidly ejected from both ventricles under high pressure.
 - **Slow Ejection Period (2nd stage, 0.09 sec):** Blood is ejected slowly with much less force.

- **End-Systolic Volume (ESV):** Amount of blood remaining in each ventricle at the end of the ejection period ($\approx 60 - 80 \text{ mL/ventricle}$).

EJECTION FRACTION (EF)

- **Definition :** Fraction (proportion) of End-Diastolic Volume that is ejected out by each ventricle per beat.
- **Formulas :**

$$EF = \frac{\text{Stroke Volume (SV)}}{\text{End-Diastolic Volume (EDV)}} = \frac{EDV - ESV}{EDV}$$

- **Normal Value :** $60 - 65\%$ (From an EDV of $130 - 150 \text{ mL}$, $\approx 70 \text{ mL}$ is ejected per beat).
- **Clinical Significance :** Best index for assessing **ventricular contractility** ($EF \downarrow$ decreases in *Myocardial Infarction* and *Cardiomyopathy*).

B. PHASES OF VENTRICULAR DIASTOLE (0.5 sec)

- **1. Protodiastole (0.04 sec) :—**
 - 1st stage of ventricular diastole.
 - Intraventricular pressure drops below aortic / pulmonary artery pressure $\longrightarrow$ **Semilunar valves close abruptly.**
 - **Second Heart Sound (S_2):** Produced by the closure of semilunar valves (aortic and pulmonary).
- **2. Isometric / Isovolumetric Relaxation Period (0.08 sec) :—**
 - Characterized by $\downarrow$ **decrease in muscular tension without change in fiber length.**
 - **All 4 valves remain closed** $\longrightarrow$ intraventricular pressure falls rapidly without volume change.
 - Fall in ventricular pressure below atrial pressure **causes AV valves to open.**
- **3. Rapid Filling Phase (0.11 sec) :—**
 - Opening of AV valves results in a **sudden rush of blood** from atria into relaxed ventricles.
 - Responsible for approximately **70% of total ventricular filling.**
 - **Third Heart Sound (S_3):** Produced by the rushing/turbulent flow of blood into the ventricles during rapid filling.
- **4. Slow Filling Phase / Diastasis (0.19 sec) :—**
 - Ventricular filling slows down as pressures equalize.
 - Responsible for approximately **20% of total ventricular filling.**
- **5. Last Rapid Filling Phase (0.11 sec) :—**
 - Coincides with **Atrial Systole.**
 - Atria contract and push the remaining small volume of blood into ventricles (**Atrial Kick**).
 - Responsible for approximately **10% of total ventricular filling.**

END-DIASTOLIC VOLUME (EDV)

- **Definition** : Total volume of blood remaining/accumulated in each ventricle at the end of diastole.
- **Normal Value** : **130 – 150 mL/ventricle.**

PRESSURE CHANGES DURING CARDIAC CYCLE

Pressure changes in the heart during the cardiac cycle are divided into **2 main categories**:

- i. **Intra-Atrial Pressure Changes**
- ii. **Intra-Ventricular Pressure Changes**

MAXIMUM AND MINIMUM PRESSURES IN CARDIAC CHAMBERS & VESSELS

Area / Chamber	Maximum Pressure	Minimum Pressure
Left Atrium	7 to 8 mmHg	0 to 2 mmHg
Right Atrium	5 to 6 mmHg	0 to 2 mmHg
Left Ventricle	120 mmHg	5 mmHg
Right Ventricle	25 mmHg	2 to 3 mmHg
Systemic Aorta	120 mmHg	80 mmHg
Pulmonary Artery	25 mmHg	7 to 8 mmHg

1. INTRA-ATRIAL PRESSURE CHANGES

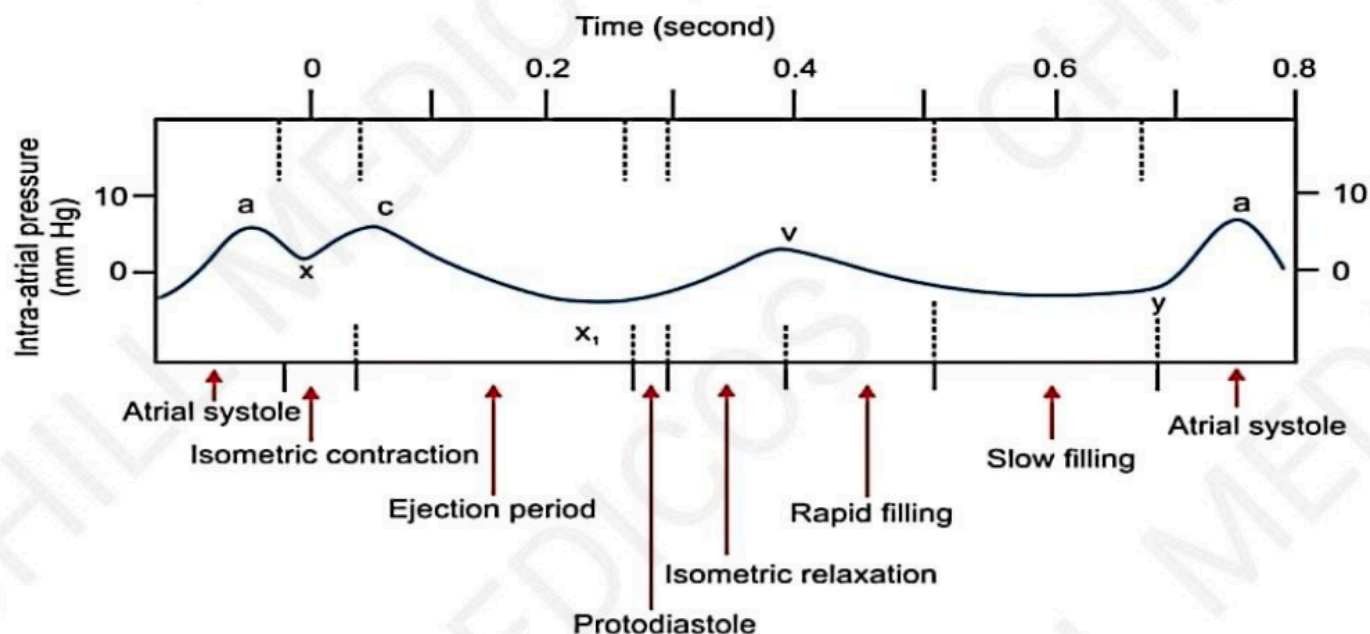


FIGURE 91.3: Intra-arterial pressure changes during cardiac cycle.
a, c, v = Positive waves. x, x₁, y = Negative waves.

- **Functional Significance :** Intra-atrial pressure is responsible for the **opening of AV valves** and **ventricular filling**, and is the main factor governing the **development of the jugular venous pulse**.
- **Phlebogram :** The intra-atrial pressure curve resembles the jugular venous pulse tracing (phlebogram).
- **Wave Composition :** Comprises **3 positive waves** (a, c, v) and **3 negative waves** (x, x₁, y).

Intra-Atrial Waves Breakdown

Wave Type	Wave	Timing / Cardiac Cycle Event	Mechanism / Cause
Positive Waves	'a' wave	Atrial Systole	Pressure rises up to 5 mmHg in Right Atrium and 7 mmHg in Left Atrium due to atrial muscular contraction.
	'c' wave	Isometric Contraction	Rising intraventricular pressure forces AV valves to bulge back into the atria , elevating intra-atrial pressure.
	'v' wave	Late Ventricular Systole (Atrial	Gradual pressure rise due to continuous venous return filling the closed atria while AV valves remain

Wave Type	Wave	Timing / Cardiac Cycle Event	Mechanism / Cause
		Diastole)	closed.
Negative Waves	'x' wave	Onset of Atrial Diastole	Due to atrial relaxation; AV valves close at the end of this wave.
	'x ₁ ' wave	Ventricular Ejection Period	Contraction of ventricular muscle pulls the atrioventricular ring downward toward the apex, expanding atrial volume and lowering pressure.
	'y' wave	Rapid Filling Phase	AV valves open → blood rushes rapidly from atria into ventricles → atrial pressure drops sharply .

2. INTRA-VENTRICULAR PRESSURE CHANGES

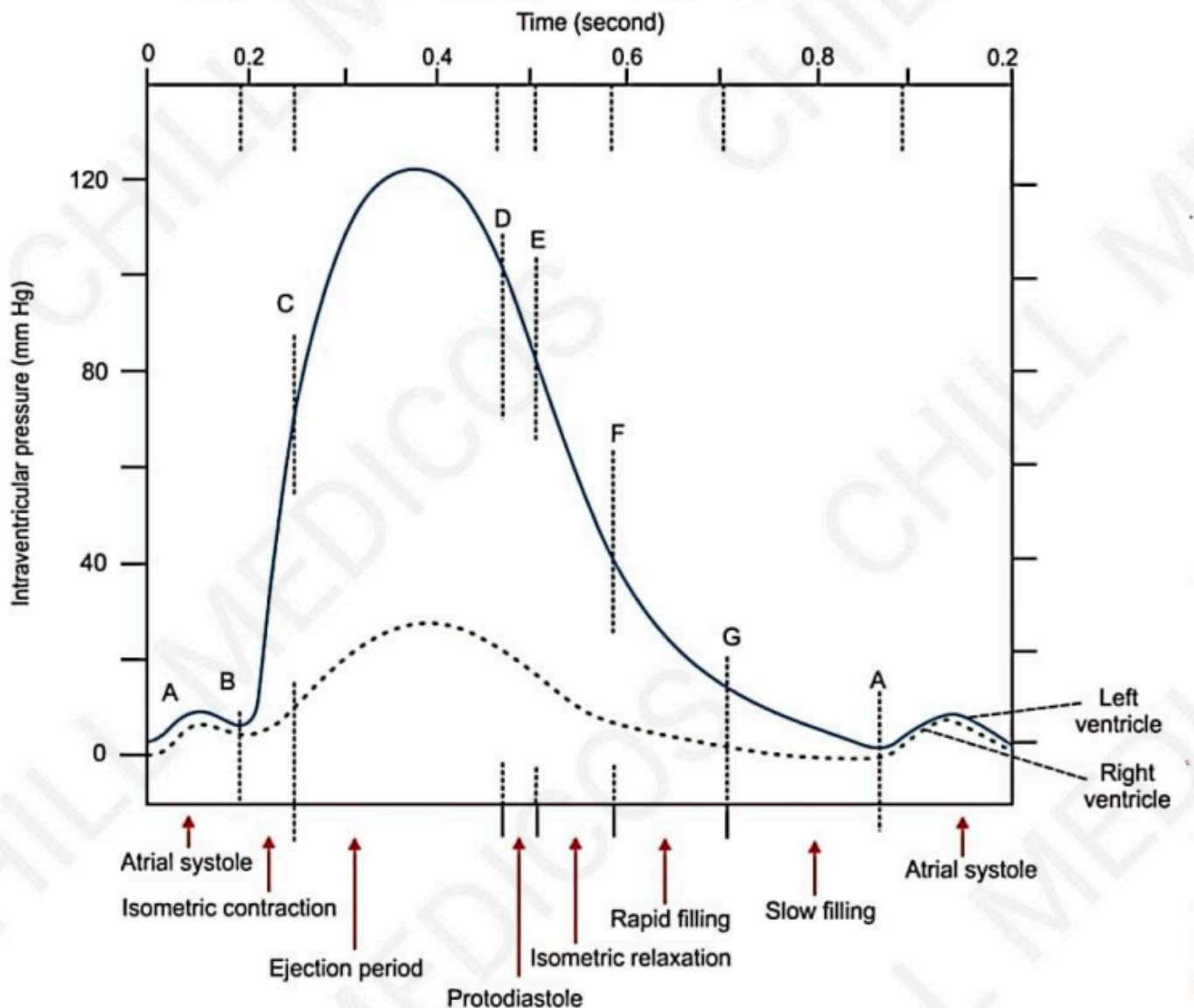


FIGURE 91.4: Intraventricular pressure changes during cardiac cycle

- **Physiological Importance** : Essential for the forward propulsion of blood through systemic and pulmonary circulations, which depends on the pressure generated during ventricular contraction.
- **Ventricular Pressure Asymmetry** : Pressure in the **Left Ventricle** (120 mmHg) is 4 – 5 times higher than in the **Right Ventricle** (25 mmHg) due to the significantly thicker muscular wall of the left ventricle.

Intra-Ventricular Pressure Curve Segments

Segment	Cardiac Cycle Phase	Pressure Changes & Physiological Events
A – B Segment	Atrial Systole	• Small volume of blood enters ventricles from atrial contraction. • Pressure rises to 6 – 7 mmHg (Right Ventricle) and 7 – 8 mmHg (Left Ventricle). • Point 'B' indicates the closure of Atrioventricular (AV) valves.
B – C Segment	Isometric Contraction Period	• Sharp, steep rise in intraventricular pressure as ventricles contract as closed cavities. • Point 'C' marks the opening of Semilunar valves when ventricular pressure exceeds arterial pressure.
C – D Segment	Ejection Period	• Maximum systolic pressure reaches 25 mmHg (Right Ventricle) and 120 mmHg (Left Ventricle). • Blood is forcefully ejected into the aorta and pulmonary artery.
D – E Segment	Protodiastole	• Pressure decreases slightly as ventricular relaxation begins. • Point 'E' indicates the closure of Semilunar valves.
E – F Segment	Isometric Relaxation Period	• Rapid, steep fall in ventricular pressure as muscle relaxes with all valves closed. • Intraventricular pressure falls below intra-atrial pressure. • Point 'F' denotes the opening of Atrioventricular (AV) valves.
F – G Segment	Rapid Filling Phase	Pressure continues to decrease/remain low despite filling due to ongoing ventricular muscle relaxation.
G – A Segment	Slow Filling Phase (Diastasis)	Continued passive filling with pressure maintained at basal resting levels before the next atrial systole.

VOLUME CHANGES DURING CARDIAC CYCLE

- **Ventricular Volume Changes :—**
 - Essential factor to maintain **cardiac output** and proper **blood circulation**.
 - The amount of blood is identical in both the **right and left ventricles**.
- **Ventricular Volume Curve** : Recorded experimentally by using a **Henderson cardiometer**.

VENTRICULAR VOLUME CURVE SEGMENTS

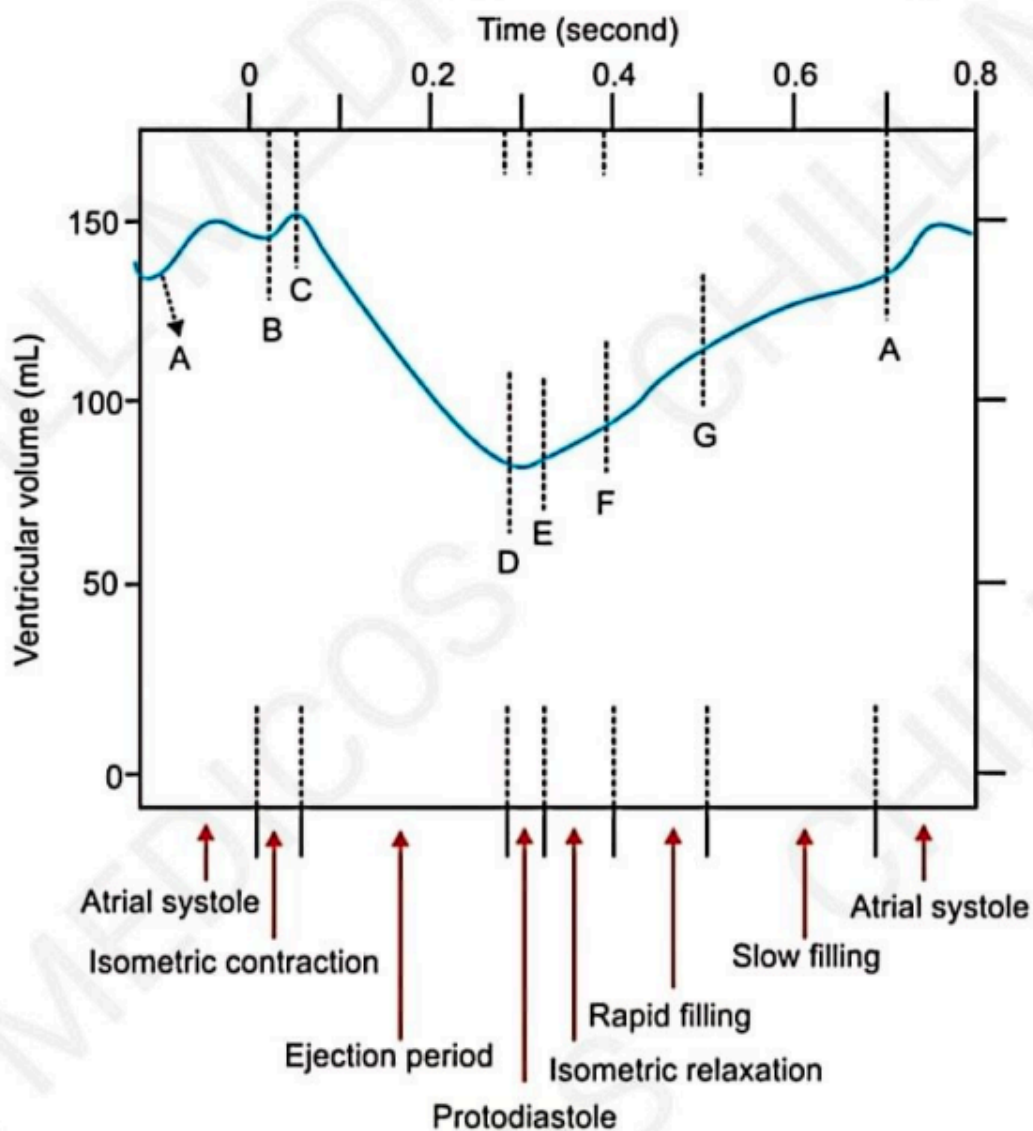


FIGURE 91.6: Ventricular volume changes during cardiac cycle

Segment	Cardiac Cycle Phase	Volume Changes & Valvular Events
1. A – B Segment	Atrial Systole (<i>Last filling phase</i>)	• A small amount of blood enters the ventricles from the atria. • Causes a slight $\uparrow$ increase in ventricular volume (reaches $EDV \approx 130 - 150 \text{ mL}$). • Point 'B' indicates the closure of Atrioventricular (AV) valves.
2. B – C Segment	Isometric Contraction Period	• Ventricular volume is not altered as the ventricles contract as closed chambers. • Any slight upward deflection seen on tracings is an artifact. • Point 'C' represents the opening of Semilunar valves.
3. C – D Segment	Ejection Period	• Rapid Ejection: Sharp, steep fall in ventricular volume. • Slow Ejection: Ventricular volume $\downarrow$ decreases slowly until it reaches $ESV (\approx 60 - 80 \text{ mL})$.
4. D – E Segment	Protodiastole	• No change in ventricular volume. • Point 'E' denotes the closure of Semilunar valves.
5. E – F Segment	Isometric Relaxation Period	• Ventricular volume is not altered. • Any slight upward deflection is an artifact caused by the entrance of blood into coronary arteries from the aorta. • Point 'F' indicates the opening of Atrioventricular (AV) valves.
6. F – G Segment	Rapid Filling Phase	• Marked rise in ventricular volume ($\approx 70\%$ of total filling). • Caused by the sudden rush of blood from the atria immediately after AV valves open.
7. G – A Segment	Slow Filling Phase (<i>Diastasis</i>)	• Ventricular volume $\uparrow$ increases slowly and steadily ($\approx 20\%$ of total filling) due to continuous venous return.

SUMMARY OF VENTRICULAR VOLUME STATUS

